



Ignored Aspects Of SMPS Design That Need Immediate Attention

When it comes to designing SMPS for electronic devices, big or small, wattages do affect as the components, layout, and topology change. As a result, the aspects to consider will also change. Paying attention to the upkeep of capacitors and related components along with resolving issues on electromagnetic interferences go a long way in tackling challenges filled with erroneous circuit challenges

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During SMPS designing, most engineers tend to overlook certain crucial aspects. After the process is complete, at first glance the design looks okay until an unstable power supply issue creates a problem in the entire product, leading to numerous complaints from the consumers. So let us take a look at these overlooked aspects and address them one by one.

The power supply is one of the most essential features of any electronic device, from lighting systems to consumer durables, or from industrial products to electric vehicles (EVs). The life of an efficient product stays in its switched mode power supply (SMPS). Designing an efficient supply not just guarantees stability in the complete circuit but also ensures the reliability of a product.

The wattage plays an important role in deciding the topology as per requirements. So, an isolated or non-isolated topology can be chosen as per the design specifications. Starting right from flyback at lower wattages to push-pull and full-bridge that are used widely in larger output requirements, many topologies exist in power electronics. But they all have the same factors for reliable output.

How to reduce capacitor failures

As per a study, about 17.1 % of all failure is due to capacitor deterioration. The major culprits behind this are ambient temperature, capacitance, and voltage stress. Commonly observed capacitance failures include:

- Swelling
- Chemical leakage (which affects nearby components)
- Capacitor explosion
- Shortening
- Reduced capacitance (which affects the overall circuit performance)

Certain preventive measures that can be taken are:

- Use high-quality capacitors
- Keep capacitors cool as much as possible
- Do not make ripple excessively stressed
- Keep in check the voltage stress factor (max. applied voltage divided by device nominal voltage)

A capacitor's shelf life should also

Selecting the Right Topology

Table 1

Input voltage	Output voltage	Preferred topology
Universal input (90-264)V AC	Po<150watt, Load current<10A	Flyback, Forward
Universal input (90-264)V AC	Po<150watt, Load current>10A	Forward
Universal input (90-264)V AC	150watt<Po>350	Two-switch Forward, Half-bridge, Push-pull
Universal input (90-264)V AC	Po<500watt	Half-bridge, Push-pull
Vin>350V DC	Po<750watt	Half-bridge
Vin<200V DC	Po<500watt	Push-pull
Vin>350V DC	500<Po>1000watt	Full-bridge
Vin>350V DC	Po>1000watt	ZVT Full-bridge
Vin>350V DC	Po>2000watt	More than one ZVT full-bridge in parallel, interleaved with more than one ZVT full-bridge